

Macro- and Microphysical Properties of Atmospheric River Snowfall: Results from Two Instrument Sites

JACK RICHTER^a, CLAIRE PETTERSEN^a, DMITRI MOISSEEV^b, JULIA SHATES^c, AND FRASER KING^a

^a *University of Michigan, Ann Arbor, Michigan*

^b *University of Helsinki, Helsinki, Finland*

^c *Jet Propulsion Laboratory, California Institute of Technology, Pasadena, California*

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ABSTRACT: Atmospheric rivers (ARs) can produce intense snowfall events with different characteristics compared to snowfall events not associated with ARs. This study uses remotely sensed and in situ observations at multiyear instrument sites in Marquette, Michigan, and Hyytiälä, Finland, to comprehensively evaluate snowfall properties during ARs and compare them to snowfall properties during non-AR conditions (NoAR). A snowfall classification scheme is applied at each site to find deep (large scale) snowfall events and classify them as either AR or NoAR snowfall. To identify differences between the two regimes, reanalysis data products and ground-based observations are utilized to examine the large-scale properties, environmental conditions, and macro- and microphysical characteristics of AR and NoAR snowfall. During AR snowfall, there is a preferential large-scale setup at both instrument sites that results in enhanced water vapor transport. The AR snowfall events produce higher snowfall rates, as well as a higher number of denser and faster-falling snow particles compared to NoAR snowfall.

SIGNIFICANCE STATEMENT: In this study, we aim to better understand the physical properties of snowfall that are coincident with an atmospheric river. This is important because atmospheric rivers produce enhanced snowfall events in mid- and high-latitude regions whose snowfall properties are relatively understudied. We find that snowfall coincident with an atmospheric river is particularly intense and produces more, faster-falling, and heavier snow particles than snowfall not coincident with an atmospheric river. Our findings also suggest that different processes may be contributing to the intense nature of atmospheric river snowfall, which can be leveraged to better understand and predict these events.

KEYWORDS: Atmospheric river; Snowfall; Instrumentation/sensors; Radars/Radar observations; Remote sensing

1. Introduction

The meridional transport of moisture is a key atmospheric process that drives mid- and high-latitude weather systems (Gimeno et al. 2016; Nash et al. 2018). Moisture transport is important to extratropical hydroclimates (Nash et al. 2018) and is especially impactful in winter months (Wypych et al. 2018). Atmospheric rivers (ARs), defined as long and narrow corridors of strong horizontal water vapor transport (AMS 2017), are extreme forms of moisture transport. Notably, ARs are responsible for the majority of the moisture transport in the midlatitudes despite being relatively small spatial features (Zhu and Newell 1998). Consequently, ARs amplify extratropical precipitation variability (Nash et al. 2018; Payne et al. 2020) and are associated with extreme winter precipitation. Lavers et al. (2011) found that the 10 largest winter flooding events in the United Kingdom were associated with ARs, and Slinkey et al. (2020) demonstrated that up to 100% of 95th percentile extreme winter precipitation events were linked to

ARs in the southeast, Midwest, and western United States. These extreme precipitation events can cause flooding, landslides, and other adverse ecological effects (e.g., Lavers et al. 2013; Gimeno et al. 2016) with economic impacts that can exceed 1 billion U.S. dollars (Corringham et al. 2019). Additionally, ARs are projected to occur more frequently in the high latitudes (Li and Ding 2024) and in eastern North America and western Europe (Lavers et al. 2013; Payne et al. 2020) based on simulations of future climate scenarios. Therefore, weather systems associated with ARs need to be characterized to better understand extreme precipitation in a changing climate.

The fraction of snowfall with respect to total precipitation in the mid- and high latitudes ranges from 10% to 90% (Kulie and Bennartz 2009; Kulie et al. 2020) and demonstrates that snowfall contributes a significant portion to total precipitation. However, future predictions of snowfall are uncertain due to climate change. As the Earth warms, snowfall events are expected to become less frequent in the Northern Hemisphere since some snowfall will shift to rainfall (Krasting et al. 2013; Danco et al. 2016); however, extreme snowfall events are predicted to become more common in high-latitude regions such as Canada, northern Europe, northern Asia, and Greenland due to increased humidity at warmer temperatures (Krasting et al. 2013; O’Gorman 2014; Danco et al. 2016; Quante et al. 2021). This predicted increase in snowfall intensity highlights

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Corresponding author: Jack Richter, rijack@umich.edu

the need to assess extreme snowfall events to improve our understanding of snowfall patterns in a warming climate.

Atmospheric rivers impact cold-season (October–April) precipitation and are associated with enhanced snow accumulation and anomalously warm temperatures (e.g., Guan et al. 2010; Gorodetskaya et al. 2014; Mateling et al. 2021; Xiao and Lettenmaier 2021; Rendfrey et al. 2024). Additional work has shown that precipitation during warm air intrusions (similar mechanism to ARs) demonstrates distinct characteristics when compared to other precipitation regime extremes, such as cold air outbreaks (e.g., Pettersen et al. 2020b; Lackner et al. 2023). Moreover, Mateling et al. (2021) found that 28% of cold-season, large-scale precipitation events within 100 km of Marquette, Michigan, were coincident with an AR, demonstrating that ARs can contribute significantly to wintertime precipitation even in regions that are far inland. While these findings reveal that ARs influence cold-season precipitation, there remains an opportunity to extensively examine if and how ARs impact the macro- and microphysical properties of snowfall. Characterizing the impact of ARs on snowfall properties is important as slight changes in these properties, such as particle sizes, concentrations, fall speeds, and densities, can lead to drastically different snow rates (e.g., Ishizaka et al. 2016; von Lerber et al. 2017; Jeoung et al. 2020), and scattering properties that affect our ability to accurately observe and forecast snowfall (e.g., Matrosov 2007; Kneifel et al. 2015). For example, one study that modeled AR precipitation found that all model configurations overpredicted total accumulation during ARs (Jankov et al. 2009). This was largely due to aggressive graupel formation that overestimated the amount of liquid accretion in the model, highlighting the importance of understanding AR snowfall so that it can be accurately predicted. Distinct microphysical properties also reflect different ice processes, such as riming and aggregation (e.g., Locatelli and Hobbs 1974; Böhm 1989; Ishizaka et al. 2013; Kneifel et al. 2015; Moisseev et al. 2017; Falconi et al. 2018), indicating that analyzing snow microphysics can provide insights into these mechanisms. By evaluating the macro- and microphysical properties of AR snowfall events, we can determine how ARs impact snow particles and lend insight into the related ice processes of AR snowfall.

2. Methodology

a. Snowfall instrument sites

We employ two multiyear instrument sites specifically configured to characterize snowfall (Fig. 1) that provide an excellent opportunity to evaluate snowfall during ARs and not during ARs (NoAR). These sites are located in Marquette, Michigan (MQT: 46.53°N, 87.55°W, 426 m MSL; Pettersen et al. 2020b; Kulie et al. 2021), and Hyttiälä, Finland (HYY: 61.844°N, 24.288°E, 150 m MSL; Petäjä et al. 2016). The instrument sites provide mostly continuous (75%–97% cold-season uptime, instrument dependent) observations of snowfall over multiple winter seasons, observe snowfall in distinct environments, and have been successfully used in previous studies of snowfall

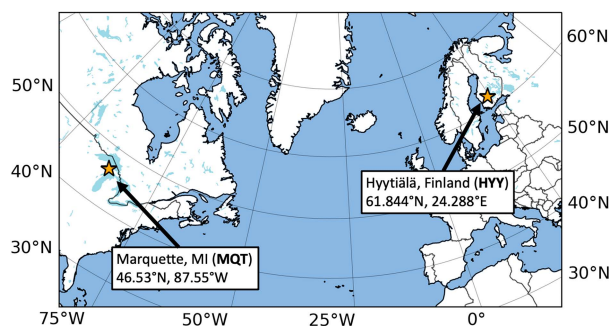


FIG. 1. Locations of the two snowfall observation sites used in this study. The MQT site is located 46.53°N, 87.55°W at 426 m MSL, and the HYY site is located 61.844°N, 24.288°E at 150 m MSL. Observations are applied from 2014 to 2022 at the MQT site and from 2017 to 2021 at the HYY site.

(e.g., Kneifel et al. 2015; von Lerber et al. 2017; Mateling et al. 2021; Shates et al. 2023; King et al. 2024b).

The MQT instrument site is collocated with a National Weather Service (NWS) Office and has been in operation since January 2014. This site is optimal for observing snowfall as it observes >200 cm of geometric snowfall accumulation annually and provides power and Internet connections (Pettersen et al. 2020b). The HYY instrument site is maintained by the University of Helsinki and the Finnish Meteorological Institute (FMI) at the Hyttiälä Forest Station. This site has a long history of conducting atmospheric research (Petäjä et al. 2016) and was established as part of the Station for Measuring Ecosystem–Atmosphere Relations research program (Hari and Kulmala 2005). In 2014, HYY was the deployment site for the Department of Energy Biogenic Aerosols—Effects on Clouds and Climate campaign (Petäjä et al. 2016), which brought additional instrumentation to the site and enabled microphysical observations of snowfall. In this study, we utilize MQT snowfall observations from January 2014 to December 2022 and HYY snowfall observations from January 2017 to December 2021 as there was similar instrumentation between the sites during these periods.

b. Instruments

1) PROFILING RADAR

Both instrument sites employ frequency-modulated continuous-wave profiling radars, which enable the detection of snowfall within the vertical column. The MQT site has a 24-GHz (K band) Metek MicroRain Radar 2 (MRR; Klugmann et al. 1996) that provides vertical profiles of reflectivity and Doppler velocity in fixed 100-m range gates up to 3000 m at 1-min temporal resolution. The MRR observations are processed using IMProToo (Maahn and Kollias 2012) to increase the sensitivity of the radar to -10 dBZ and are “de-noised” using a principal component analysis method on the raw power spectra to remove interference (Pettersen et al. 2020b). We discard MRR observations from the first three range gates due to ground clutter, leaving an operating range of 400–3000 m above ground level (AGL). The MRR has been used in numerous studies examining both

snowfall and rain (e.g., Kneifel et al. 2011; Pettersen et al. 2020b; Schoger et al. 2021; Shates et al. 2021, 2023), demonstrating its utility as a precipitation radar.

The HYY site has a 94-GHz (W band) Radiometer Physics GmbH radar (RPG-94; Kuchler et al. 2017; Moiseev and Petäjä 2024). The RPG-94 is configured in range gates of about 30 m at approximately 3-s temporal resolution. To roughly match the range resolution of the MRR, we chunk the RPG-94 range gates every four bins to get a range resolution of 89–136 m (dependent on height AGL). We similarly discard RPG-94 observations below 400 m and above 3000 m to be consistent with the MRR, leaving a range of 400–3000 m AGL. A water vapor attenuation correction is applied to the RPG-94 data. While W-band radars were originally designed for the observation of clouds, previous studies have reinforced the utility of using them in studies of snowfall (e.g., Matrosov et al. 2008; Falconi et al. 2018). Furthermore, the RPG-94 has been used to study snow microphysics (e.g., Jeoung et al. 2020; Billault-Roux et al. 2023) and in the development of a snowfall rate retrieval (Schoger et al. 2021).

The profiling radars serve two primary purposes. The first is to examine the time–height reflectivity at each site to discriminate between deep and shallow precipitation when defining snowfall events of interest [see section 2d(1)]. The second purpose is to evaluate aggregated radar profiles of AR and NoAR snowfall at each site to assess differences in reflectivity characteristics during the regimes using both height AGL and temperature as a vertical coordinate.

2) PRECIPITATION IMAGING PACKAGE

Both sites host a Precipitation Imaging Package (PIP; Newman et al. 2009; Pettersen et al. 2020a; King et al. 2024a), a National Aeronautics and Space Administration (NASA)-developed video disdrometer that takes high-speed videos of falling hydrometeors. The PIP's field of view is 64 mm × 48 mm with a pixel resolution of 0.1 mm and is aimed at a halogen lamp such that falling particles cast shadows in the high-speed videos. These shadows are analyzed using image processing techniques to create particle size distributions (PSDs) and fall speed distributions at 1-min resolution (Pettersen et al. 2020a). The PIP additionally uses empirically derived parameterizations to determine the bulk effective density, effective density distributions, and phase-separated liquid water equivalent (LWE) precipitation rates (e.g., Pettersen et al. 2021; King et al. 2024a). The effective density values are computed using inputs of the PSD and fall speed distribution, so they inherently covary with those observations (Pettersen et al. 2020a).

The PIP observations are leveraged in multiple ways herein. The first is to utilize the PIP bulk effective density measurements to discriminate between liquid and frozen precipitation as part of the snowfall detection methodology [see section 2d(1)]. The second is to aggregate the PIP observations and measurements to create composite two-dimensional (2D) PSD, fall speed, and effective density datasets of AR and NoAR snowfall at each site so that the microphysical properties of the regimes can be compared. We also integrate the PSDs to get the total number of snow particles (N_i) during the events.

The PIP PSD observations are additionally used to empirically fit 15-min running average PSDs to the gamma size distribution for snow particles:

$$N(D) = N_0 D^\mu \exp(-\lambda D), \quad (1)$$

where $N(D)$ is the number of particles ($\text{m}^{-3} \text{mm}^{-1}$), D is the particle diameter (mm), N_0 is the intercept parameter ($\text{m}^{-3} \text{mm}^{-1}$), λ is the slope parameter (mm^{-1}), and μ is the shape factor (e.g., Ulbrich and Atlas 1998; Tiira et al. 2016). We fit the gamma PSDs to examine the characteristic PSD shapes for the AR and NoAR snowfall events. While Eq. (1) is not the only way to fit PSDs (e.g., Gunn and Marshall 1958; Testud et al. 2001; Heymsfield et al. 2008), it is a commonly used parameterization that we employ for comparisons.

3) SURFACE METEOROLOGICAL MEASUREMENTS

The surface meteorological measurements at the MQT site are obtained from the automated Davis Vantage Pro2 weather station (Davis Instruments 2018; King et al. 2024a), which is mounted on a tower (10 m AGL) adjacent to the NWS office. Surface temperature and wind speed measurements at MQT are used in this study and are reported in 5-min intervals. The surface meteorological data at the HYY site are obtained from the Juupajoki Hyytiälä weather station, WMO no. 2770, which is located 200 m from the cloud radar (Moiseev and Petäjä 2024). Measurements of surface temperature and wind speed at HYY are used and are reported at approximately 3-s temporal resolution.

c. Reanalysis

We implement the fifth generation European Centre for Medium-Range Weather Forecasts (ECMWF) atmospheric reanalysis (ERA5; Hersbach et al. 2020) hourly data products on single levels (Copernicus Climate Change Service 2018a) to characterize the large-scale properties of the AR and NoAR snowfall events. Specifically, we examine mean sea level pressure (MSLP), 10-m u component of wind, 10-m v component of wind, vertical integral of northward water vapor flux, and vertical integral of eastward water vapor flux from 2014 to 2022 in MQT and from 2017 to 2021 in HYY. We also use ERA5 hourly data on pressure levels (Copernicus Climate Change Service 2018b) to obtain profiles of temperature and specific humidity for each regime, and to convert to geopotential heights so that we can examine the radar characteristics at both sites using temperature as the vertical coordinate (O'Connor 2024). The wind and water vapor flux vector components are used to calculate wind speeds and integrated water vapor transport (IVT) magnitudes. The ERA5 products have a horizontal resolution of $0.25^\circ \times 0.25^\circ$, and the profiles have 37 pressure levels from 1000 to 1 hPa. This study utilizes the ERA5 grid points closest to MQT: 46.5°N , 87.5°W and HYY: 61.75°N , 24.25°E for compositing the vertical profiles at each site. Last, we use ERA5 monthly averaged data (Hersbach et al. 2023) from 1979 to 2019 to create monthly climatological values of MSLP and temperature so that we can evaluate the large-scale anomalies of MSLP and temperature at both sites.

d. Snowfall classification scheme

To evaluate the characteristics of AR and NoAR snowfall, we first develop a snowfall classification scheme to identify and categorize events of interest. The scheme is composed of two conditions: a deep (i.e., large scale) snow detection and an AR presence near the site. If both conditions are met for an event, we classify it as an AR snowfall event. If only the deep snow detection is met, we classify it as a NoAR snowfall event. Thus, the only difference between AR and NoAR snowfall events is the presence of an AR, allowing us to characterize the impacts of ARs on synoptically forced snowfall.

1) DEEP SNOW DETECTION ALGORITHM

The first component of our snowfall classification scheme is the deep snow detection algorithm. This algorithm was specifically developed for this study and examines the vertical radar reflectivity and PIP effective density measurements to identify events where there are ≥ 2 h of continuous deep, synoptically forced snowfall. We define deep snowfall as events that have precipitation echo-top heights ≥ 3 km (Pettersen et al. 2020b) and near-surface (400 m AGL) reflectivity values ≥ -10 dBZ (minimum sensitivity of the MRR). Precipitation echo-top heights are defined as the highest radar profile bin where reflectivity is ≥ -10 dBZ. The height threshold is included to purposely omit instances of shallow snowfall since shallow snowfall often occurs due to mesoscale processes such as lake- and sea-effect snowfall and can impact both sites (e.g., Kristovich 1993; Niziol et al. 1995; Mazon et al. 2015; Olsson et al. 2023). To verify that an event is producing snow as opposed to rain, we apply a PIP bulk effective density threshold of 0.4 g cm^{-3} to ensure mostly solid precipitation (Pettersen et al. 2021; King et al. 2024a). Overlapping periods of precipitation echo-top heights ≥ 3 km and effective density $\leq 0.4 \text{ g cm}^{-3}$ are classified as deep (i.e., large scale) snowfall events if the length of the overlap is ≥ 2 h and count as independent events (for statistical purposes) if the events are at least 12 h separated from other events.

2) AR DETECTION ALGORITHM

To identify the occurrence of ARs, we apply the Mattingly AR Detection Algorithm (Mattingly et al. 2018; hereafter M18). The M18 algorithm has been successfully utilized in studies of midlatitude (e.g., Mateling et al. 2021) and high-latitude ARs (e.g., Mattingly et al. 2023; Zhang et al. 2023; Mateling et al. 2024). The M18 algorithm applies two IVT thresholds that both need to be satisfied in order for an AR detection: an IVT magnitude threshold of $150 \text{ kg m}^{-1} \text{ s}^{-1}$ and an 85th percentile IVT threshold. The 85th percentile IVT threshold in the M18 algorithm is reanalysis-based and evolves temporally and spatially at each grid point on a ± 15 day running window (see details in M18). These are relatively low IVT thresholds compared to other AR detection algorithms that are specific to lower-latitude regions (e.g., Leung and Qian 2009; Rutz et al. 2014; Gershunov et al. 2017) but are consistent with other AR detection algorithms that have thresholds for high-latitude regions (e.g., Lavers et al. 2012; Guan and Waliser 2015; Ramos et al. 2016).

Lower IVT thresholds ensure that enhanced moisture transport events in relatively cold and dry regions, as is the case with our instrument sites, are identified as ARs. Additional geometry requirements are applied by the M18 algorithm to only include ARs that are >1500 km in length and have a length-to-width ratio > 1.5 (M18). Precipitation associated with an AR typically occurs within 100 km of the AR major axis (e.g., Nayak et al. 2016); therefore, we include all ARs detected within 100-km radius of the sites. To ensure that any snowfall being influenced by an AR is properly classified, any deep snowfall on a calendar day with an AR detection is classified as AR snowfall. Furthermore, radar observations are manually inspected when there is a calendar day transition to verify that the AR snowfall events are continuous.

Applying the M18 algorithm to the entire Northern Hemisphere during cold-season months (October–April) from 2014 to 2022 is shown in Fig. 2. For this period, ARs were most frequent over the Atlantic and Pacific Oceans and were least frequent over northern North America, northern Asia, and in regions of complex topography. Importantly, ARs occurred at a frequency $\geq 10\%$ at both the MQT and HYY instrument sites, highlighting that these sites provide a great opportunity to study AR snowfall.

An example in which the snowfall classification scheme identified an AR snowfall event at the MQT instrument site is portrayed in Fig. 3. The height-reflectivity time series from the MRR profiling radar for this event shows deep (i.e., large scale) precipitation occurring from 0200 to 1200 UTC and is marked by the magenta horizontal line (Fig. 3a). The average effective density is $\leq 0.4 \text{ g cm}^{-3}$ from 0700 to 1900 UTC and is marked by the orange horizontal line (Fig. 3b). The overlap of the magenta and orange horizontal lines between 0700 and 1200 UTC (cyan box, Figs. 3a,b) indicates when a deep snowfall event occurred during this time frame. Figure 3c outlines an AR that was detected by the M18 algorithm at 1200 UTC on the same date. Since the MQT site is within a 100 km of the AR, we classify the deep snowfall on this day as AR snowfall. If instead there was not an AR detected within a 100 km of the site on this day, the deep snowfall would be classified as NoAR snowfall.

e. Statistical testing

To test the significance between the AR and NoAR snowfall properties, we implement a difference of means test:

$$z = \frac{\bar{x}_1 - \bar{x}_2}{\left(\frac{s_1^2}{N_1} + \frac{s_2^2}{N_2}\right)^{0.5}}, \quad (2)$$

where $\bar{x}$ is the sample mean, s is the sample standard deviation, and N is the effective sample size (Wilks 2011). For this study, the z statistic is used to calculate the p value and determine if a given difference of the means between the AR and NoAR snowfall events is statistically significant at the 95% confidence level. The effective sample size is determined by counting the number of independent AR and NoAR snowfall events at each site. This is done to reduce overconfidence of the test statistic in Eq. (2) as the events may otherwise be correlated. The effective

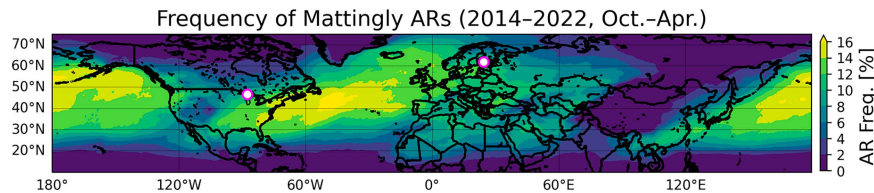


FIG. 2. Frequency of ARs during the cold season (defined as October–April) using the Mattingly AR detection scheme (M18) from 2014 to 2022. The frequency of AR occurrence for a location is expressed as a percentage of the total number of time steps during the study period and filled according to the color bar. The two white dots with magenta outlines depict the locations of the instrument sites with the MQT site at 46.53°N, 87.55°W and the HYY site at 61.844°N, 24.288°E.

sample sizes of the regimes at each site are the number of events reported in Table 1.

3. Results

We apply the snowfall classification scheme described in section 2 to identify AR and NoAR snowfall events. Table 1 lists the number of AR and NoAR snowfall events that were detected at the sites between 2014–22 (MQT) and 2017–21 (HYY), their average duration, and the standard deviation of their durations. At MQT, there were 35 AR snowfall events and 105 NoAR snowfall events, while at HYY, there were 36 AR snowfall events and 70 NoAR snowfall events. Individual AR snowfall events lasted 8.47 and 7.18 h on average at MQT and HYY, respectively, and the NoAR snowfall events lasted 5.60 and 10.24 h on average at MQT and HYY, respectively.

Key surface and near-surface properties of the AR and NoAR snowfall events at each site are included in Table 2. The properties listed are the mean values for a given site and regime. At both sites, the AR snowfall events have higher mean surface temperatures, mean surface wind speeds, LWI snow rates, near-surface reflectivity, and near-surface Doppler velocities than the NoAR snowfall events. Several of these differences are statistically significant using a difference of means test (see bolded figures in Table 2).

Using these datasets of AR and NoAR snowfall events at each site, we first evaluate the large-scale properties to understand the synoptic conditions and environmental profiles that correspond to each regime. After discerning the large-scale properties, we examine the macro- and microphysical properties to determine the snowfall characteristics of each regime. By exploring these properties from large scale to microscale, we can determine the impacts that ARs have on snowfall and connect them to snowfall processes.

a. Large-scale properties

1) MQT

The ERA5 hourly products are composited and averaged for the AR and NoAR snowfall events at each site while the events are ongoing to evaluate the characteristic large-scale and environmental properties of the regimes. Composites showing the large-scale conditions at the MQT site for the AR and NoAR snowfall events are shown in Fig. 4 and include MSLP (Figs. 4a,b) and IVT (Figs. 4e,f). Both the AR and NoAR

snowfall events show a region of low pressure over the Upper Great Lakes region (Figs. 4a,b). Notably, there is a distinct difference in the locations of high pressure between the two regimes. High pressure > 1022 hPa is located over the eastern United States during the AR snowfall events (Fig. 4a), whereas high pressure > 1022 hPa is located to the northwest of the site over Alberta and Saskatchewan, Canada, during the NoAR snowfall events (Fig. 4b). As such, the composite 10-m winds during the AR snowfall events show strong northward flow from the Gulf of Mexico and southern United States, whereas the 10 m composite winds during the NoAR snowfall events show stronger southeastward flow from the Great Plains and central Canada.

The IVT magnitude composites for these regimes reveal that during AR snowfall events, there is $IVT > 150 \text{ kg m}^{-1} \text{ s}^{-1}$ near the MQT instrument site (Fig. 4e), whereas during the NoAR snowfall events (Fig. 4f), there is $IVT < 100 \text{ kg m}^{-1} \text{ s}^{-1}$. This enhanced IVT is a requirement of the AR detection algorithm, so this is not surprising and rather confirms the performance of the M18 algorithm. The predominant source of moisture for both the AR and NoAR snowfall events appears to be the Gulf of Mexico since a large region of $IVT > 250 \text{ kg m}^{-1} \text{ s}^{-1}$ extends from the Gulf to the upper Midwest during both regimes, and $IVT < 100 \text{ kg m}^{-1} \text{ s}^{-1}$ is found elsewhere. However, this moisture from the Gulf extends further north and adjacent to the MQT site during the AR events.

Additionally shown in Fig. 4 are the composite MSLP anomalies (Figs. 4c,d) and 2-m temperature anomalies (Figs. 4g,h) for each regime at the MQT site. The anomalies are calculated by subtracting the relevant monthly climatological values (1979–2019) from the reanalysis data and averaging over all time steps to get the composite anomalies. The white dots on the anomaly plots (Figs. 4c,d,g,h) indicate where the anomalies are statistically significant with respect to climatology at the 95% confidence level using a difference of means test [Eq. (2)]. Both the AR and NoAR snowfall events in MQT show statistically significant low pressure composite anomalies in the Great Lakes region (Figs. 4c,d), with the AR events having greater magnitude low pressure composite anomalies than the NoAR events (-8 hPa vs -6 hPa). Notably, the locations of high pressure composite anomalies differ between the two regimes. During the AR events, there is a large area of statistically significant high pressure composite anomalies east of the MQT site that peaks around $+7$ hPa (Fig. 4c). While the NoAR events also have an area of statistically significant high

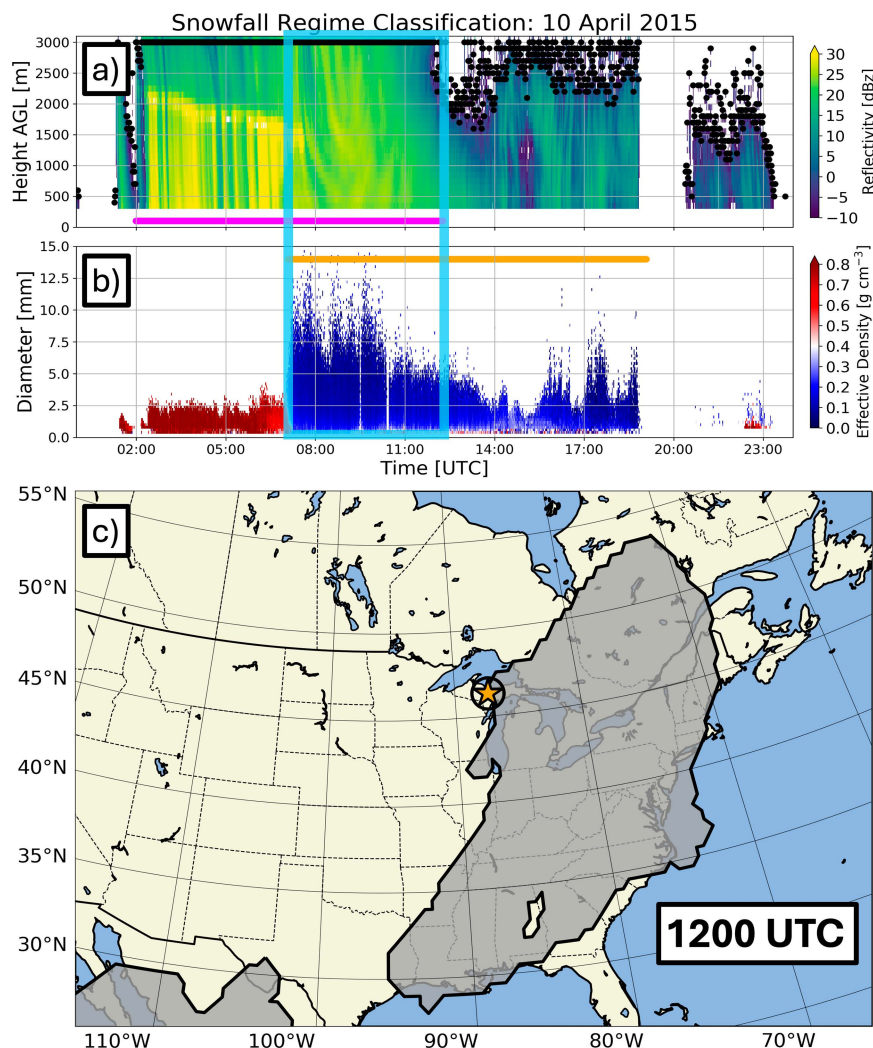


FIG. 3. AR snowfall event detection at the MQT site on 10 Apr 2015. (a) The profiling radar reflectivity (color filled), precipitation echo-top heights (black markers), and deep precipitation flag (magenta horizontal line) as a function of time for the day. (b) The PIP effective density distribution (color filled) and occurrence of the frozen precipitation flag (orange horizontal line) as a function of time for the day. (c) The location and extent of the AR for the given date at 1200 UTC (black line and gray fill; MQT site location is marked by the orange star with a 100-km radial ring in black). Where the magenta and orange horizontal lines intersect [cyan box; (a) and (b)] defines the deep snowfall event and since an AR was also detected within 100 km of the site during the event (c), it is classified as an AR snowfall event.

pressure composite anomalies east of the MQT site, it is only +4 hPa, and there is also a region of +5-hPa high pressure composite anomalies northwest of the MQT site (Fig. 4d). For these reasons, the anomalous 10-m winds from the Gulf of Mexico and Atlantic Ocean are of a higher magnitude during the AR events (Fig. 4c) than during the NoAR events (Fig. 4d).

The higher-magnitude 10-m wind anomalies from the Gulf of Mexico and Atlantic Ocean during the AR events contribute to higher-composite 2-m temperature anomalies during the AR events than the NoAR events (Figs. 4g,h). The AR snowfall events show statistically significant warm temperature anomalies

stretching from the southern United States through central Canada, with 2°–3°C anomalies shown near the MQT site (Fig. 4g). Conversely, the NoAR snowfall events show near 0°C composite temperature anomalies near the MQT site and statistically significant composite cold temperature anomalies across Canada and the northern Great Plains (Fig. 4h).

2) HYY

Correspondingly, the large-scale composites and anomalies are shown for the AR and NoAR snowfall events at the HYY site (Fig. 5). The two regimes have similar MSLP composites

TABLE 1. The number, average duration, and event duration standard deviation (stdev) of AR and NoAR snowfall events detected at each of the instrument sites using the snowfall classification scheme.

	MQT		HYY	
	AR snow	NoAR snow	AR snow	NoAR snow
Number of events	35	105	36	70
Avg. event duration (h)	8.47	5.60	7.18	10.24
Event duration stdev (h)	5.48	2.70	4.36	6.17

and are characterized by a broad region of low pressure in the northern Atlantic Ocean and a broad region of high pressure northwest of Africa, south of the low pressure region. However, the magnitude of the composite low pressure region is different for the two regimes. For the AR snowfall events (Fig. 5a), the low pressure region has a minimum low of 996 hPa, whereas for the NoAR snowfall events (Fig. 5b), the low pressure minimum is 1004 hPa. This >8 hPa difference corresponds to a stronger pressure gradient during the AR snowfall events—the isobars are closer together, and there are stronger composite winds than the NoAR snowfall events. This stronger pressure gradient extends all the way to HYY during the AR snowfall events, injecting a lobe of low pressure ~1000 hPa near the site (Fig. 5a). Comparatively, there is a weaker pressure gradient near HYY during the NoAR snowfall events and low pressure only ~1004 hPa near the site (Fig. 5b).

The AR snowfall events in HYY show IVT > 150 kg m⁻¹ s⁻¹ reaching the instrument site (Fig. 5e), whereas the NoAR snowfall events in HYY show IVT < 100 kg m⁻¹ s⁻¹ (Fig. 5f). This enhanced IVT during the AR snowfall events closely follows the region of tight isobars and strong 10-m winds shown in Fig. 5a. Moreover, the northern Atlantic Ocean appears to be the primary source of moisture, as there is a broad region of IVT > 250 kg m⁻¹ s⁻¹ in the northern Atlantic during both regimes. This enhanced IVT from the Atlantic extends further east toward the HYY site during the AR events.

Additionally shown in Fig. 5 are the composite MSLP anomalies (Figs. 5c,d) and 2-m temperature anomalies (Figs. 5g,h) for

each regime. Both the AR and NoAR snow events in HYY show statistically significant (with respect to climatology) low pressure composite anomalies in northern Europe (Figs. 5c,d), with the AR events having greater magnitude low pressure composite anomalies than the NoAR events (−10 hPa vs −8 hPa). Notably, the area of composite low pressure anomalies covers a larger region during the AR events than the NoAR events. During the AR events, the low pressure composite anomalies extend west over the North Atlantic, whereas the low pressure composite anomalies only cover northern Europe during the NoAR events (Figs. 5c,d). Consequently, the 10-m wind composite anomalies from the North Atlantic are of a higher magnitude during the AR events (Fig. 5c) than NoAR events (Fig. 5d), inducing enhanced moisture transport across the North Atlantic toward the HYY site during AR snowfall.

The higher-magnitude 10-m wind anomalies from the North Atlantic during the AR events contribute to higher-composite 2-m temperature anomalies during the AR events than the NoAR events (Figs. 5g,h). During the AR snowfall events, there is a large region with statistically significant warm temperature anomalies over Europe, with 3°–4°C anomalies shown near the HYY site (Fig. 5g). Conversely, the NoAR snowfall events show near 0°C composite temperature anomalies near the HYY site, with statistically significant negative composite temperature anomalies to the west of the HYY site over the Scandinavian Peninsula (Fig. 5h).

3) VERTICAL PROFILES

To connect the synoptic-scale features to the environments at each site, composite profiles of ERA5 temperature and specific humidity are examined (Fig. 6). At the MQT site, the AR snowfall events have significantly (95% confidence level) warmer mean temperatures throughout the atmospheric column compared to the NoAR snowfall events (Fig. 6a). This difference is about 2°–3°C in the lower atmosphere (1000–850 hPa) and is about 3°–5°C in the middle atmosphere (850–500 hPa). The AR snowfall events also have significantly higher-mean specific humidities throughout the atmospheric column compared to the NoAR snowfall events at MQT (Fig. 6b). This difference is about 0.4–0.5 g kg⁻¹ in the lower atmosphere and is about 0.5–0.7 g kg⁻¹ in the middle atmosphere. At the HYY site, the AR snowfall events also have significantly higher-mean

TABLE 2. Summary of environmental and bulk microphysical properties at the two instrument sites (MQT and HYY) for the AR and NoAR snowfall events. All values reported are the mean of the regime at the site (the standard deviations of the values are provided in parentheses). The standard deviations of the near-sfc reflectivity values are provided in linear units since dBZ is logarithmic (asterisked parentheses). The near-sfc radar properties are obtained by selecting the first height bin above 400 m AGL at each site. Abbreviations: sfc = surface; LWE = liquid water equivalent; V = velocity; stdev = standard deviation. Bolded text indicates that the differences between AR and NoAR mean values are significant at the 95% confidence level (*p* values also included).

	MQT			HYY		
	AR snow	NoAR snow	<i>p</i> value	AR snow	NoAR snow	<i>p</i> value
Sfc temperature (°C)	−3.694 (3.603)	−5.447 (5.146)	0.026	−2.246 (2.674)	−5.272 (4.580)	<0.001
Sfc wind speed (m s ^{−1})	3.290 (1.769)	2.173 (1.453)	<0.001	6.520 (3.564)	4.416 (2.596)	0.002
LWE rate (mm h ^{−1})	1.269 (1.486)	0.745 (0.955)	0.051	0.579 (0.571)	0.330 (0.427)	0.021
Near-sfc Reff (dBZ)*Stdev (mm ⁶ m ^{−3})	17.428 (65.276)*	14.917 (39.724)*	0.038	7.918 (6.400)*	6.413 (5.039)*	0.139
Near-sfc Doppler V (m s ^{−1})	1.317 (0.730)	1.190 (0.523)	0.341	1.118 (0.703)	0.950 (0.514)	0.204

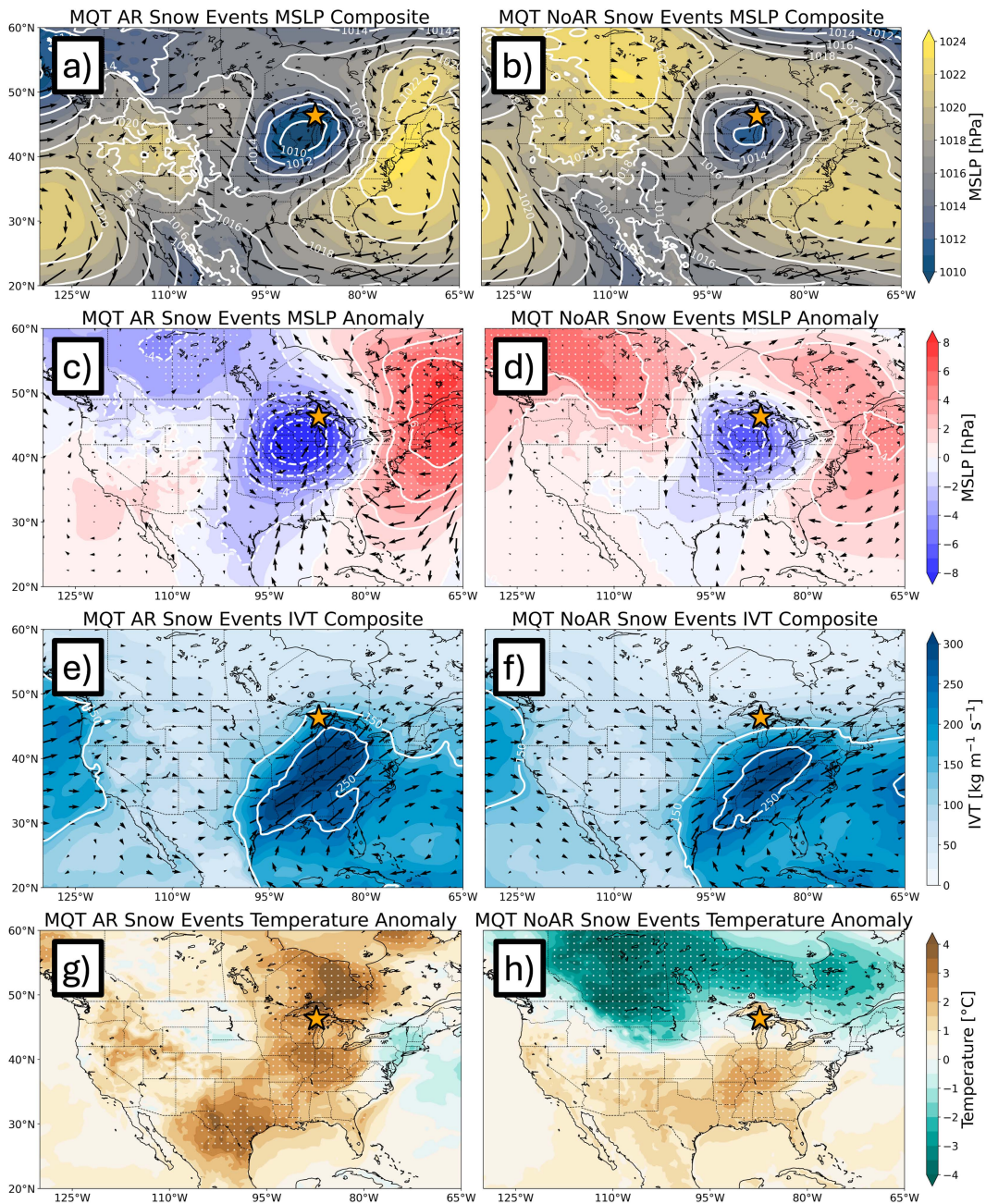


FIG. 4. (a),(b) ERA5 MSLP composites, (c),(d) MSLP composite anomalies, (e),(f) IVT composites, and (g),(h) 2-m temperature composite anomalies for the (left) AR snowfall events and (right) NoAR snowfall events at the MQT instrument site. MSLP is color filled and contoured in hectopascals on (a)–(d), with 10-m winds depicted by the black arrows. IVT magnitude is color filled and contoured at 150 and 250 $\text{kg m}^{-1} \text{s}^{-1}$ on (e) and (f), with IVT vectors depicted by the black arrows on (e) and (f). Temperature is color filled in degrees Celsius on (g) and (h). The location of the MQT instrument site is marked by the orange star. The white dots on the anomaly plots (c), (d), (g) and (h) indicate where the anomalies are statistically significant with respect to climatology at the 95% confidence level using a difference of means test.

temperatures and specific humidities throughout the atmospheric column compared to the NoAR snowfall events (Figs. 6c,d). These differences are similarly about $2^{\circ}\text{--}3^{\circ}\text{C}$ and $0.4\text{--}0.5 \text{ g kg}^{-1}$ in the lower atmosphere, and about $3^{\circ}\text{--}5^{\circ}\text{C}$ and $0.5\text{--}0.7 \text{ g kg}^{-1}$ in the middle atmosphere.

b. Macro- and microphysical properties

1) RADAR COMPOSITES

Radar reflectivity profiles at each site are aggregated and normalized into 2D histograms for the AR and NoAR

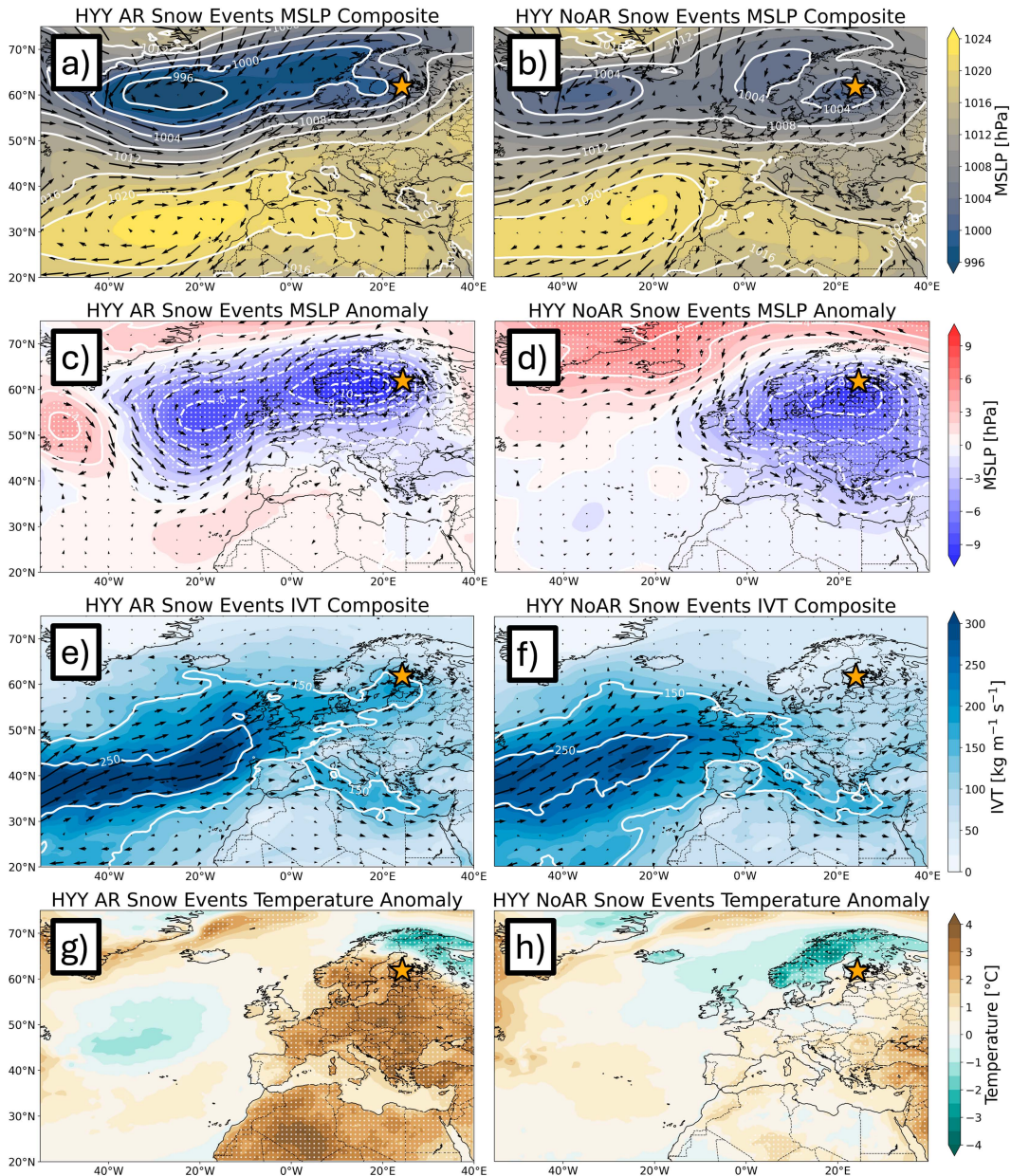


FIG. 5. As in Fig. 4, except for the events at the HYY instrument site. The location of the HYY instrument site is marked by the orange star.

snowfall events to assess differences in reflectivity characteristics between the regimes. These normalized 2D reflectivity histograms are included for the AR (Figs. 7a,d) and NoAR (Figs. 7b,e) snowfall events in MQT (top row) and HYY (bottom row). At both sites, the AR snowfall events have reflectivity histograms (Figs. 7a,d) that are shifted to higher reflectivity compared to the NoAR events (Figs. 7b,e). The reflectivity histogram distributions look similar between the two sites, with the HYY histograms shifted about 10-dBZ lower compared to the MQT site, likely due to attenuation.

The normalized 2D reflectivity histograms for the AR and NoAR snowfall events at each site are subtracted from each

other to reveal radar reflectivity properties that are characteristic to each regime. These histograms are included for MQT (Fig. 7c) and HYY (Fig. 7f). Positive regions (red) highlight radar reflectivity characteristics that are more commonly observed during AR snowfall and negative regions (blue) highlight those that are more commonly observed during NoAR snowfall. The MQT 2D normalized difference reflectivity histogram (Fig. 7c) shows positive regions at higher reflectivity values than the negative regions for all heights AGL. This reveals that AR snowfall has higher reflectivity values throughout the column than NoAR snowfall in MQT. As such, the mean near-surface (400 m AGL) reflectivity value during the AR snowfall events

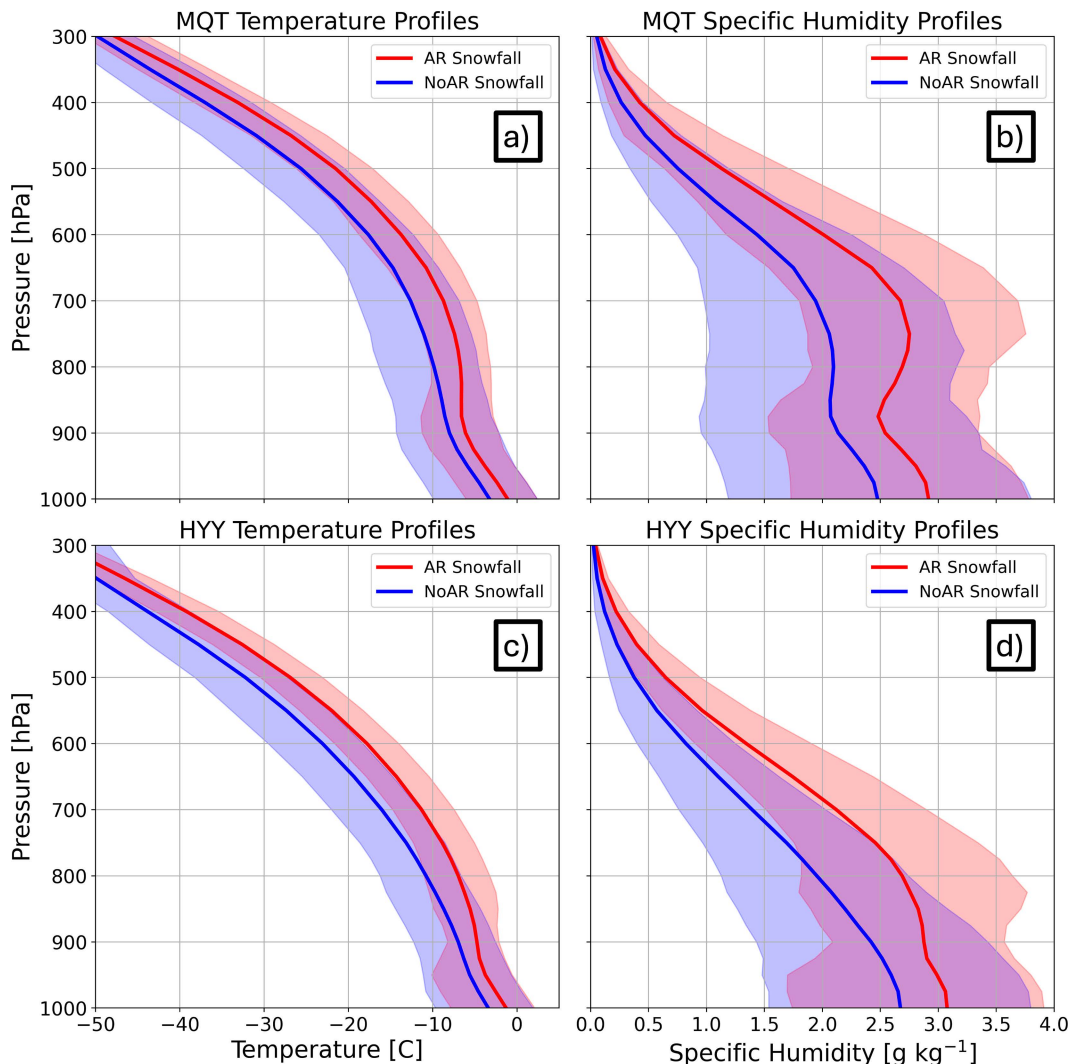


FIG. 6. (a),(c) Composites of ERA5 temperature and (b),(d) specific humidity profiles for the AR snowfall events (red) and NoAR snowfall events (blue) in (top) MQT and (bottom) HYY. The solid lines represent the mean value at each pressure level, and the envelopes fill between the 10th and 90th percentiles at each pressure level. For MQT, we use ERA5 data at 87.5°W, 46.5°N and for HYY, we use ERA5 data at 61.75°N, 24.25°E. The differences between the mean temperature and specific humidity profiles are significant at both sites from 1000 to 300 hPa. The ERA5 temperature profiles are also shown in skew T -log p diagrams to assess the individual profiles (Fig. S1).

is 2.51 dBZ greater than the NoAR snowfall events in MQT (Table 2). Additionally, the mean near-surface Doppler velocity is 0.13 m s^{-1} greater during the AR snowfall events in MQT, indicating that the AR snowfall events have faster falling particles.

The HYY normalized difference reflectivity histogram shows lower reflectivity magnitudes compared to the MQT histogram because of attenuation effects and potential non-Rayleigh scattering at W band. That being said, the HYY 2D normalized difference reflectivity histogram (Fig. 7f) also shows positive regions at higher reflectivity values than the negative regions, revealing that AR snowfall also has higher reflectivity values throughout the column in HYY. Accordingly, the mean near-surface reflectivity value during the AR snowfall events is 1.51 dBZ greater than the NoAR snowfall

events in HYY (Table 2). The mean near-surface Doppler velocity is 0.17 m s^{-1} greater during the AR snowfall events in HYY, also indicating that the AR snowfall events have faster falling particles.

To further examine the different radar characteristics during AR and NoAR snowfall, 2D normalized temperature versus reflectivity histograms are shown for the AR and NoAR snowfall events in MQT and HYY (Fig. S2 in the online supplemental material). At both sites, the AR snowfall events (Figs. S2a,d) demonstrate a shift to warmer temperatures and higher reflectivity values compared to the NoAR snowfall events (Figs. S2b,e). These differences are accentuated in Figs. S2c and S2f, which show the subtracted normalized temperature versus reflectivity histograms at MQT and HYY, respectively. The differenced

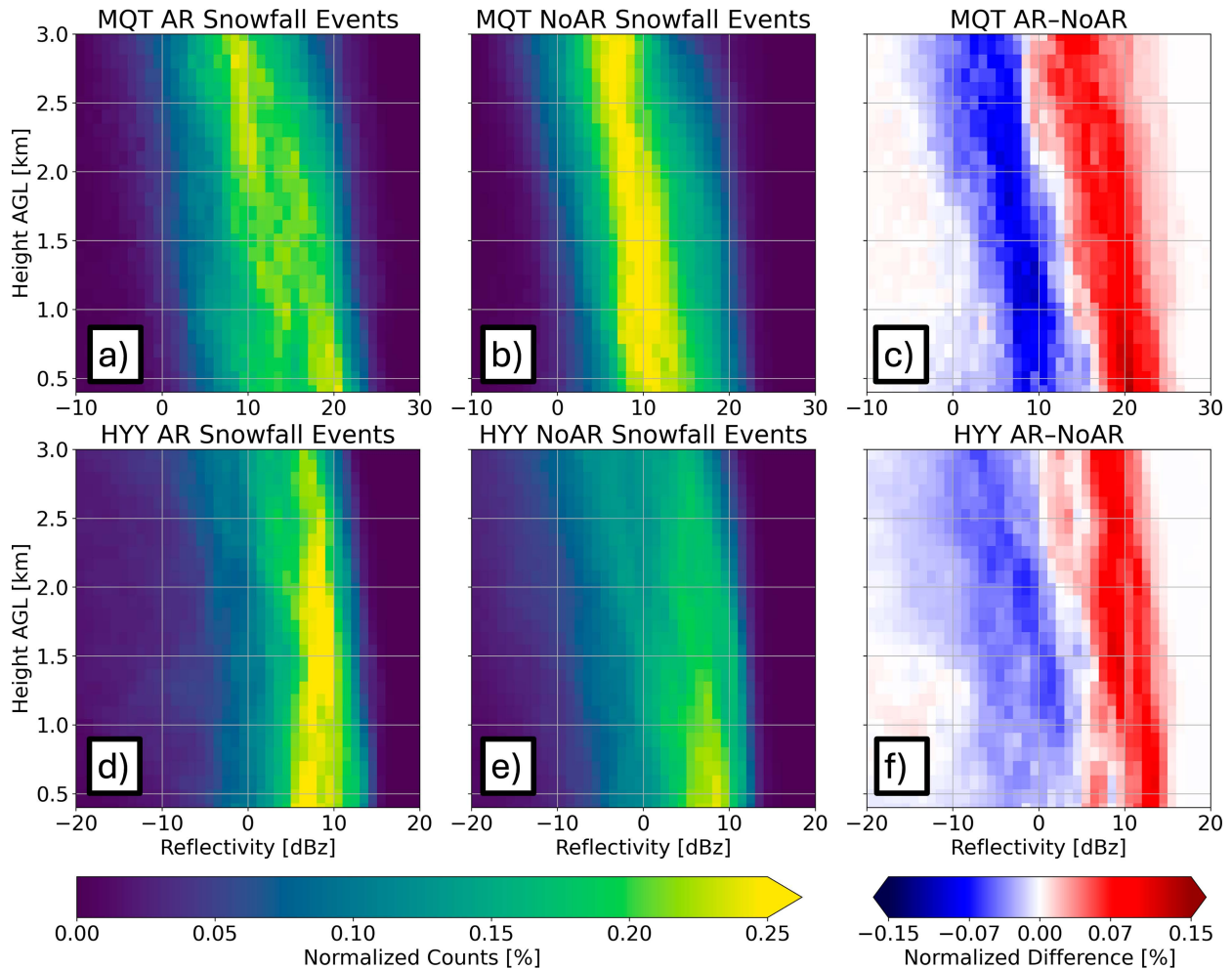


FIG. 7. (a),(d) Composite 2D radar reflectivity histograms for the AR snowfall events and (b),(e) NoAR snowfall events (top) at the MQT instrument site and (bottom) at the HYY instrument site. Each histogram is normalized by total counts and color filled according to the color bar. (c),(f) Additionally shown are the differenced reflectivity histograms at each site, with the positive values highlighting characteristics more common during the AR snow events. The bin widths are 1 dBZ for reflectivity and approximately 100 m for height above ground level (AGL). The RPG-94 (HYY) is limited to 3 km AGL to match the MRR (MQT). Attenuation is likely present >15 dBZ for the RPG-94 (HYY). Note that the reflectivity distributions would look different at the two sites even if identical snowfall events were sampled due to frequency-dependent scattering properties at W band vs K band.

histograms both show positive values at warm temperatures and high reflectivity, demonstrating that the AR snowfall events are shifted to warmer temperatures and higher reflectivity values compared to the NoAR snowfall events (Figs. S2c,f).

2) PIP COMPOSITES

The PIP observes falling snow particles, providing PSDs and fall speed distributions every minute during the AR and NoAR snowfall events. These distributions are composited and normalized at each site to compare the microphysical properties of AR and NoAR snowfall. Composite normalized 2D PIP PSD histograms are shown for the AR and NoAR snowfall events at each site (Fig. 8). At MQT, the shape of the PSD histogram and distribution of normalized counts for the AR snowfall events (Fig. 8a) is similar to the PSD histogram for the NoAR snowfall events (Fig. 8b). Both these histograms show higher

concentrations ($>0.35\%$) at diameters < 2 mm and PSD values between 10^2 and 10^4 $\text{m}^{-3} \text{mm}^{-1}$, and lower concentrations for particles >2 mm. The normalized differenced PSD histogram, the median, and 10th and 90th percentile PSDs for the two regimes in MQT (Fig. 8c) accentuate the differences between the AR and NoAR snowfall PSDs. Figure 8c shows higher PSD concentrations at each diameter for the AR snowfall events than the NoAR snowfall events, indicating that AR snowfall produces more particles than NoAR snowfall (note: the PSDs are shown in Log_{10}). Integrating the difference between the two median PSDs in Fig. 8c informs that there is a 38% increase [about 600 more particles $\text{m}^{-3} \text{min}^{-1}$ (min^{-1} is the temporal resolution of the PIP)] in the number of particles during AR snowfall compared to NoAR snowfall.

At the HYY site, the shape of the PSD histograms and distribution of normalized counts for the AR snowfall events

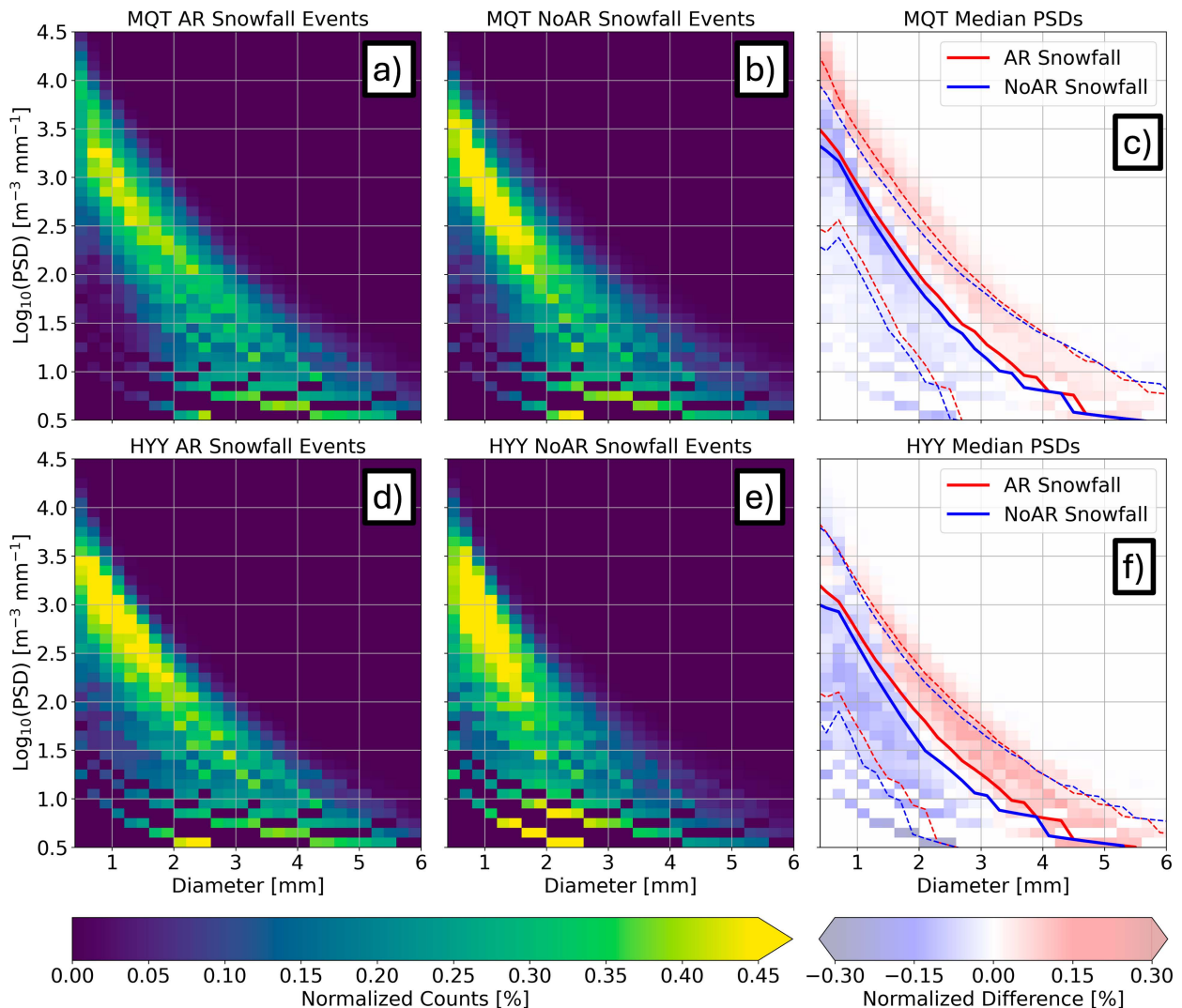


FIG. 8. (a),(d) Composite 2D PIP PSD histograms for the AR snowfall events and (b),(e) NoAR snowfall events (top) at the MQT instrument site and (bottom) at the HYY instrument site. Each histogram is normalized by total counts and color filled according to the color bar. The normalized differenced PSDs at each site are shown in (c),(f), with the positive values highlighting characteristics more common during the AR snow events. Additionally shown in (c) and (f) are the median PSDs at each site [(c) and (f); solid lines] and the 10th and 90th percentile PSDs at each site [(c) and (f); dashed lines] for each event regime (red: AR, blue: NoAR). The bin widths are 0.2 mm for diameter and $0.1 \text{ m}^{-3} \text{ mm}^{-1}$ in $\log_{10}$ scale for particle counts. The discretization of the PIP measured values of particle counts and diameter is due to the finite measurement volume of the PIP.

(Fig. 8d) is also similar to the PSD histogram for the NoAR snowfall events (Fig. 8e). Both these histograms show higher concentrations ($>0.40\%$) at diameters $< 2 \text{ mm}$ and PSD values between 10^2 and $10^{3.5} \text{ m}^{-3} \text{ mm}^{-1}$, and lower concentrations for particles $> 2 \text{ mm}$. The normalized differenced PSD histogram, the median, and 10th and 90th percentile PSDs also show higher PSD values at all diameters for the AR snowfall events compared to the NoAR snowfall events (Fig. 8f). Integrating the difference between the two median PSDs in Fig. 8f results in a 47% increase [about 400 more particles $\text{m}^{-3} \text{ min}^{-1}$ (min^{-1} is the temporal resolution of the PIP)] in the number of particles for AR snowfall compared to NoAR snowfall.

Composite normalized 2D fall speed histograms are shown in Fig. 9 and show the relative instances of particle fall speeds as a function of diameter for the AR (Figs. 9a,d) and NoAR (Figs. 9b,e) snowfall events at each site. Typical fall speed relations from Locatelli and Hobbs (1974) and Ishizaka et al. (2013) for various snowfall categories are also plotted on the histograms. The fall speed histograms at the MQT and HYY sites show similar occurrences of particle fall speeds for the AR and NoAR snowfall events. While these histograms are similar, the NoAR snowfall events are mostly concentrated below the “densely rimed” relation (Figs. 9b,e), whereas the AR snowfall events have more counts above the densely rimed relation (Figs. 9a,d).

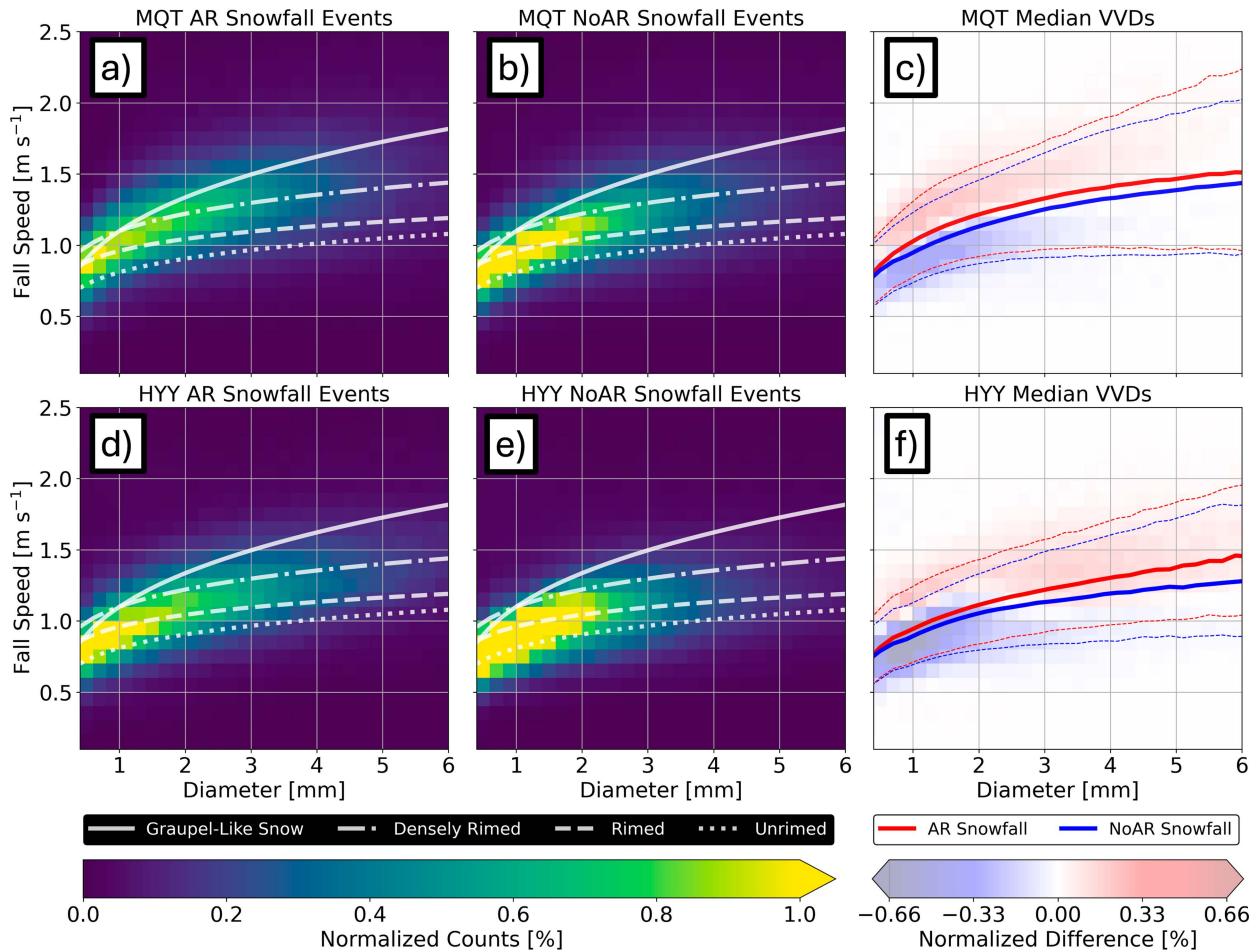


FIG. 9. As in Fig. 8, except for the 2D PIP fall speed distribution histograms. The histograms (a),(b),(d),(e) are normalized by total counts and color filled according to the color bar. The differenced fall speed distributions at each site are shown in (c),(f), with the positive values highlighting fall speed characteristics that are more common during the AR snow events. Additionally shown in (c) and (f) are the median fall speed distributions at each site [(c) and (f); solid lines] and the 10th and 90th percentile distributions at each site [(c) and (f); dashed lines] for each event regime (red: AR, blue: NoAR). The bin widths are 0.2 mm for diameter and 0.1 m s⁻¹ for fall speed. Typical fall speed relations for various snowfall categories from Locatelli and Hobbs (1974) and Ishizaka et al. (2013) are plotted on top of the fall speed histograms (a), (b), (d) and (e).

To further examine the particle fall speeds, the median, and 10th and 90th percentile fall speed distributions are shown over the normalized differenced histograms for the AR and NoAR snowfall events at MQT (Fig. 9c) and HYY (Fig. 9f). For these distributions, the AR snowfall events at both sites show faster falling particles than the NoAR snowfall events at all diameters. The differences between the median fall speed distributions indicate that the AR snowfall events have particles that fall an average of 0.077 and 0.094 m s⁻¹ faster per size bin than the NoAR snowfall events in MQT and HYY, respectively. These faster fall speeds imply that the AR snowfall events have denser particles than the NoAR snowfall events. This is confirmed by examining the effective density distributions at each site (Fig. S3), which reveal that the AR snowfall events have particles that are about 0.01–0.02 g cm⁻³ denser per size bin than the NoAR snowfall events (Figs. S3c,f).

c. PSD parameterization

To demonstrate how PSD parameters relate to other snowfall properties, hexbin-style plots depicting the occurrence of the gamma PSD parameters from Eq. (1)— N_0 and λ —are included for the AR and NoAR snowfall events at MQT (Fig. 10) and HYY (Fig. 11). The AR snowfall events in MQT have the most occurrences of N_0 values between 10 000–50 000 m⁻³ mm⁻¹ and λ values that center around 2–4 mm⁻¹ (Fig. 10a), whereas the NoAR snowfall events in MQT have the most occurrences of N_0 values between 5000–100 000 m⁻³ mm⁻¹ and λ values 3–7 mm⁻¹ (Fig. 10b). At the HYY site, the AR snowfall events have the most occurrences of N_0 values between 2000–100 000 m⁻³ mm⁻¹ and λ values between 1.5 and 5 mm⁻¹ (Fig. 11a), while the NoAR snowfall events in HYY have the most occurrences of N_0 values between 20 000–200 000 m⁻³ mm⁻¹ and λ values 4–7 mm⁻¹ (Fig. 11b). The subtracted differences

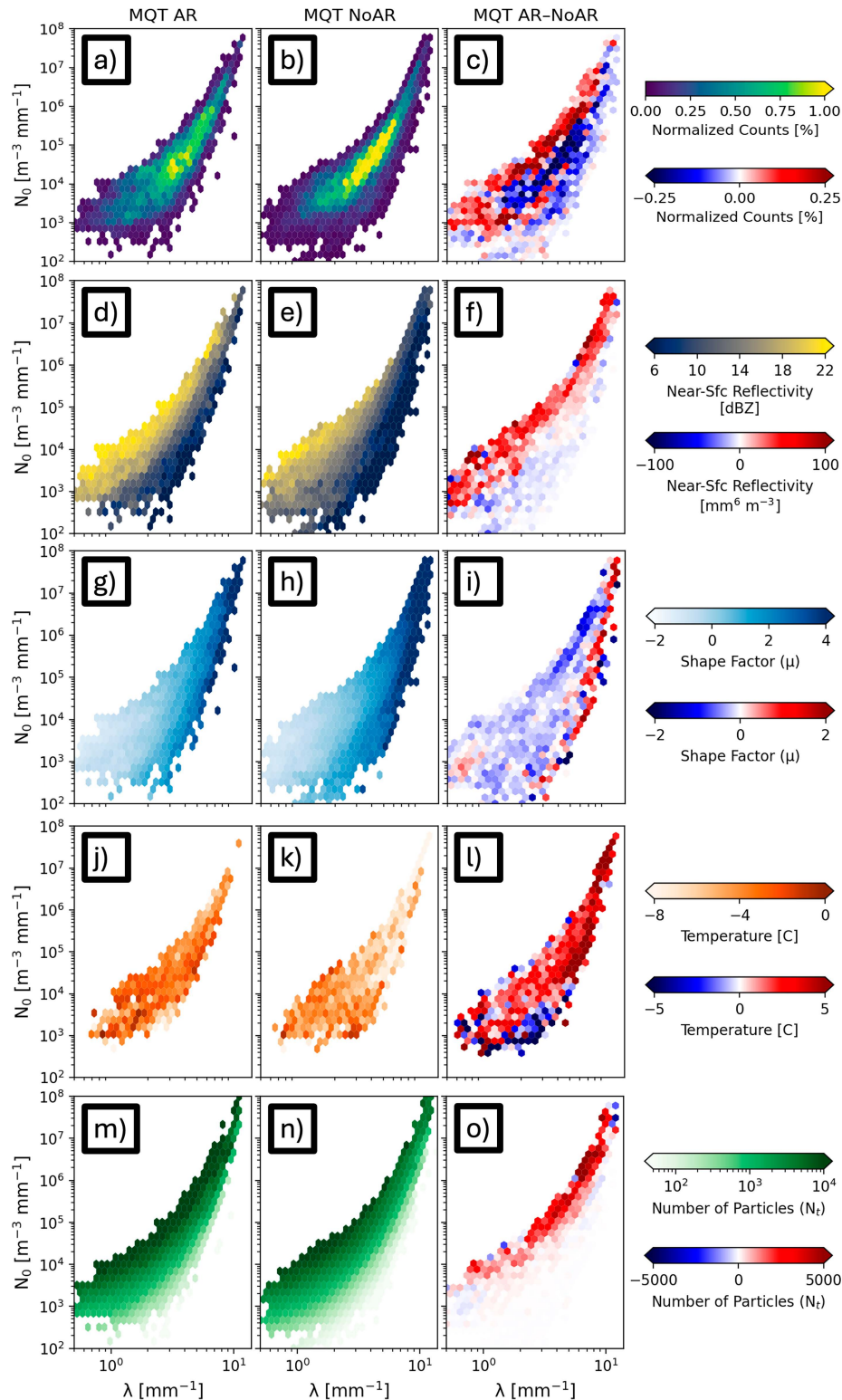


FIG. 10. Hexbin-style plots depicting the occurrence of intercept (N_0) and slope (λ) parameters from fitted gamma PSDs at the MQT site for the (a) AR and (b) NoAR snowfall events. Also included are plots that relate the coincident mean (d),(e) near-sfc reflectivity, (g),(h) shape factor (μ), (j),(k) surface temperature, and (m),(n) number of particles (N_t) in N_0 - λ space for the AR and NoAR snowfall

between the regimes are also shown in these hexbin-style plots (Figs. 10c and 11c). These differences show that the AR snowfall events tend to occupy a distinct region in the gamma PSD N_0 - λ space at each site compared to the NoAR snowfall events. For example, the AR events tend to have more instances of high N_0 and low λ combinations than the NoAR events at both sites (Figs. 10c and 11c).

The subtracted differences can be leveraged further to identify key differences in near-surface properties between the regimes. Additionally shown in Figs. 10 and 11 are the mean near-surface (sfc) reflectivity values (d, e), shape factors (μ ; g, h), surface temperatures (j, k), and N_t (m, n) in N_0 - λ space. For a given N_0 , low λ values appear to represent the most intense snowfall in this study as these regions correspond with high-mean near-sfc reflectivity, high N_t , and is more prevalent during AR snowfall than NoAR snowfall (Figs. 10c and 11c). Low λ values also correspond with near-zero μ values (Figs. 10g,h and 11g,h), and the differenced shape factors (Figs. 10i and 11i) demonstrate that AR snowfall events exhibit PSDs that are more exponential [i.e., the D^μ term in Eq. (1) reduces to 1] than the NoAR snowfall events. Subtracting individual N_0 - λ pairs between the AR and NoAR snowfall events reveals that the AR events generally have higher coincident mean near-surface reflectivity (Figs. 10f and 11f) and higher N_t (Figs. 10o and 11o) for the same pair. Both sites also show that the AR snowfall events have warmer mean surface temperatures in this N_0 - λ space (Figs. 10l and 11l) than the NoAR snowfall events.

4. Discussion

a. Large-scale and environmental analysis

The distinct large-scale and environmental characteristics of AR and NoAR snowfall at both sites indicate that there are specific synoptic conditions that lead to AR snowfall. In MQT, MSLP lag composites reveal that the AR and NoAR snowfall events both approached the Upper Great Lakes region from the south (Fig. S4), which is consistent with previous studies of deep (large scale) winter precipitation in MQT (Kulie et al. 2021; Mateling et al. 2021; Pettersen et al. 2020b). Additionally, MSLP (Fig. S4) and IVT (Fig. S5) lag composites during the AR snowfall events in MQT are similar to those found for all AR precipitation across the entire cold season in Mateling et al. (2021).

By assessing the MSLP and IVT results together from Fig. 4, we deduce that AR snowfall events in MQT are characterized by moderate low pressure over the Upper Great Lakes that combines with very high pressure in the east to funnel enhanced moisture from the Gulf of Mexico and southern United States toward the site. This results in enhanced IVT at the MQT site (Fig. 4e), which is a condition of ARs. Conversely, the NoAR snowfall events in MQT are characterized by weaker low pressure over the Upper Great Lakes that combines with high

pressure to the northwest to funnel comparatively cold and dry air from Canada toward the MQT site.

Composite MSLP differences between the AR and NoAR snowfall events in MQT (Fig. S6a) reveal that the main difference in the large-scale setups between the two regimes is the location and magnitude of composite high pressure. High pressure (>1022 hPa) is found east of the Upper Great Lakes during the AR snowfall events and is found northwest of the Upper Great Lakes during the NoAR snowfall events (Figs. 4a,b). This difference could indicate the preferential presence of an atmospheric block over the eastern United States during AR snowfall events. Atmospheric blocks are relatively stationary systems that act to slow down and modify the large-scale flow (Berggren et al. 1949; Rex 1950). Blocks can affect local precipitation and environmental characteristics by inducing anomalous moisture and warm air advection (Masato et al. 2014; Henderson et al. 2016; Breeden et al. 2020) and have been associated with enhanced snowfall in Greenland (Pettersen et al. 2022) and Japan (Yamazaki et al. 2015). During the AR snowfall events at MQT, the western side of the composite high pressure over the eastern United States corresponds with higher moisture transport (as indicated by the IVT shown in Fig. 4e) into the Upper Great Lakes region. The implication of warm air advection during the AR snowfall events at MQT is also reflected in the composite ERA5 temperature anomalies (Fig. 4g), where there are significantly warmer temperatures near the MQT site compared to the NoAR events (Fig. 4h).

Figure 5 shows that AR snowfall events in HYY coincide with strong low pressure (<996 hPa) over the northern Atlantic Ocean coupled with high pressure (>1022 hPa) to the northwest of Africa that creates a tight pressure gradient and strong 10-m winds. These winds transport moisture from the northern Atlantic Ocean toward the HYY site and result in enhanced IVT (Fig. 5e). Conversely, the NoAR snowfall events in HYY are characterized by weaker low pressure over the northern Atlantic Ocean, resulting in weaker 10-m winds and lower IVT near the site. MSLP lag composites for the AR and NoAR snowfall events in HYY (Fig. S7) show that these synoptic setups are relatively persistent in the 48 h preceding and 24 h succeeding the events. The persistent tight pressure gradient over the North Atlantic that precedes the AR snowfall events appears to push high IVT near the HYY site compared to the NoAR snowfall events (Fig. S8).

Composite MSLP differences between the AR and NoAR snowfall events in HYY (Fig. S6b) reveal that the main difference in the large-scale setups between the AR and NoAR snowfall events at HYY is the magnitude of the composite MSLP gradient in the northern Atlantic Ocean. Whereas the spatial distributions of low and high pressure were the same for the AR and NoAR snowfall events (Figs. 5a,b), the magnitude of the pressure gradient is about 8 hPa larger during the AR snowfall

←
events, respectively. The hexbins are color filled according to the color bars. The third column in each row (c), (f), (i), (l), (o) shows the subtracted differences between the AR and NoAR events for each quantity. Red regions correspond to values that are higher during AR events, whereas the blue regions correspond to values that are higher during NoAR events.

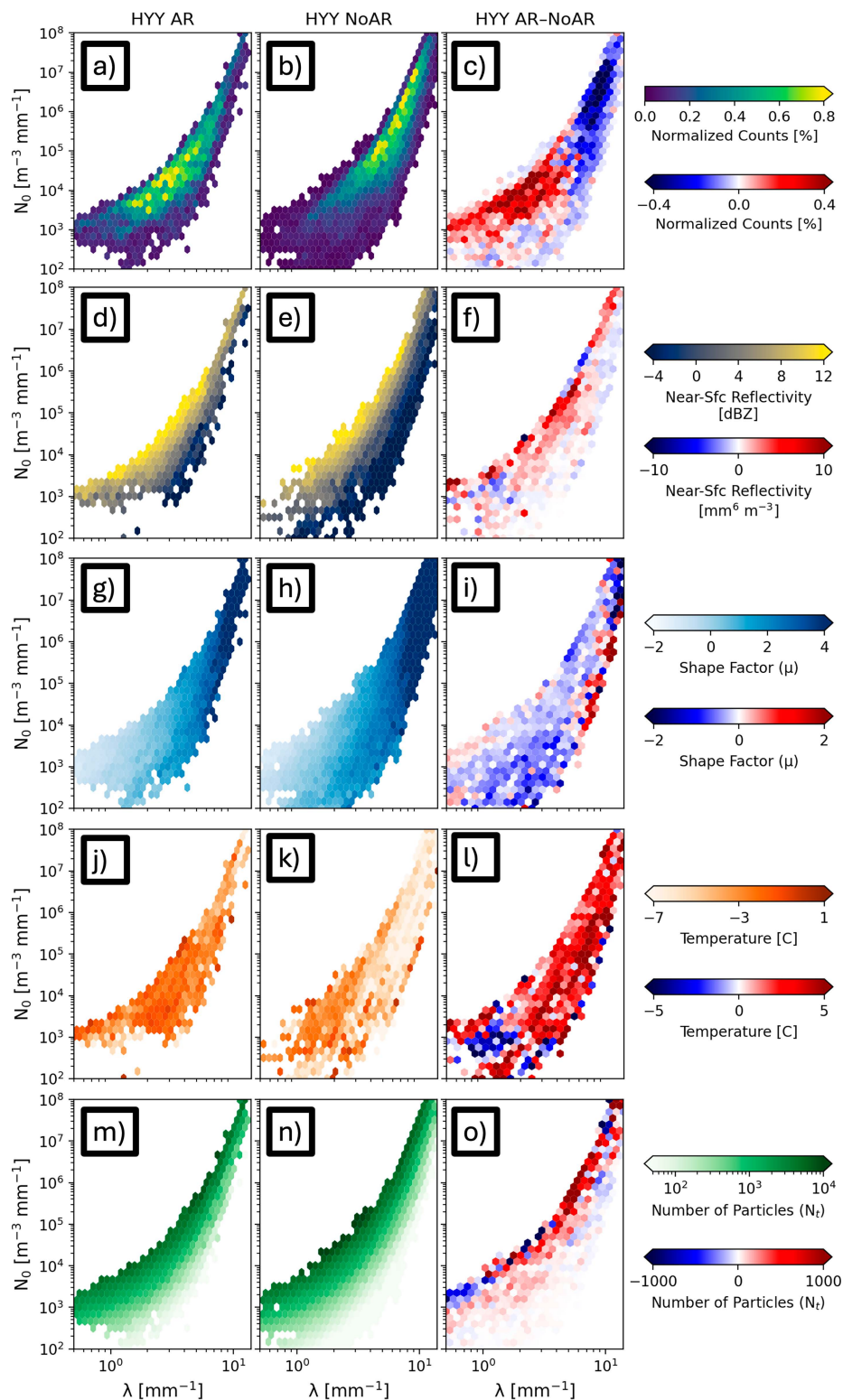


FIG. 11. As in Fig. 10, except for the snowfall events at the HYY site.

events than the NoAR snowfall events. The low and high pressure setups for the two regimes are typical of common large-scale cyclonic weather patterns discussed in [Pasquier et al. \(2019\)](#). The AR snowfall large-scale setup in HYY is typical of an Atlantic Trough pattern, whereas the NoAR snowfall large-scale setup is typical of a zonal pattern (see [Pasquier et al. \(2019\)](#)). Both these weather patterns correspond with higher extreme precipitation frequency in Finland, with the Atlantic Trough pattern having more enhancement. Coincidentally, [Pasquier et al. \(2019\)](#) also determined that the Atlantic Trough pattern had the highest frequency of AR occurrence in Finland, highlighting this connection between common large-scale weather patterns and ARs. Another large-scale mechanism that may contribute to AR snowfall in Finland is North Atlantic Jet (NAJ) extensions. [Winters and Attard \(2022\)](#) showed that tight pressure gradients such as the one exhibited during the AR snowfall events in HYY ([Fig. 5a](#)) are characteristic of NAJ extensions. These NAJ extensions bring warm temperatures to Europe ([Winters and Attard 2022](#)), which is reflected in the composite ERA5 temperature anomalies ([Figs. 5g,h](#)). During the AR snowfall events, significantly warm temperatures anomalies are shown across Europe ([Fig. 5g](#)), whereas neutral and cool temperature anomalies are shown during the NoAR snowfall events ([Fig. 5h](#)).

b. Relating microphysical properties to physical processes

The 2D normalized difference reflectivity histograms ([Figs. 7c,f](#)) demonstrate that the AR snowfall events have higher reflectivity values throughout the column than the NoAR snowfall events at each site. In addition, the mean near-surface reflectivity value during the AR snowfall events is 2.51 dBZ higher than the NoAR snowfall events in MQT and is 1.51 dBZ higher in HYY ([Table 2](#)). These higher reflectivities indicate that AR snowfall events produce more intense snowfall (e.g., [Atlas et al. 1995](#); [Matrosov 2007](#); [Kneifel et al. 2011](#); [Falconi et al. 2018](#)) than NoAR snowfall events. The mean AR LWE snow rates are also 70% higher than the NoAR LWE snow rates at both sites ([Table 2](#)), further highlighting this intense nature.

To understand why the AR snowfall events produce more intense snowfall, we examine the median PIP PSDs, fall speed, and effective density distributions. The median PIP PSDs at each site ([Figs. 8c,f](#)) show a shift to higher number concentrations for the AR snowfall events versus the NoAR snowfall events and correspond to higher reflectivity values ([Atlas et al. 1995](#)). The mean near-surface Doppler velocities ([Table 2](#)) and median PIP fall speed distributions at each site ([Figs. 9c,f](#)) indicate that the AR snowfall events have particles that fall faster than the NoAR snowfall events. These higher mean Doppler velocities and fall speeds imply that the AR snowfall events have denser particles and higher reflectivity values ([Böhm 1989](#); [Atlas et al. 1995](#); [Matrosov et al. 2002](#)). This is confirmed by evaluating the median PIP effective density distributions at each site ([Figs. S3c,f](#)), which show denser particles across the size distribution during the AR snowfall events at both sites. These microphysical differences explain why the ARs produce more intense snowfall: Denser and more plentiful particles

directly lead to more intense snow rates and, consequently, higher reflectivity values during the AR snowfall events.

To connect these outcomes to potential ice processes, we first examine the median PSDs at both sites ([Figs. 8c,f](#)). The median AR snowfall PSDs exhibit flattening (i.e., less steep slope) between 1.5 and 3.0 mm compared to the median NoAR snowfall PSDs. PSD flattening is associated with aggregation (e.g., [Westbrook et al. 2004](#)), suggesting that aggregation may be more common during AR snowfall. Next, we inspect the median fall speed distributions ([Figs. 9c,f](#)) and compare them to empirically derived fall speed relations for different ice habits. The median fall speed distributions for the AR snowfall events at both sites show fall speeds around 0.75 m s^{-1} at the smallest size bin that gradually increase to about 1.5 m s^{-1} at the 6-mm size bin. The median fall speed distributions for the NoAR snowfall events are similar but show slightly slower fall speeds at all particle sizes. Both these fall speed distributions are similar to empirically derived fall speed relations for rimed particles ([Böhm 1989](#); [Ishizaka et al. 2013](#)), with the AR snowfall distributions suggesting more densely rimed particles ([Ishizaka et al. 2013](#)). Densely rimed particles are associated with faster fall speeds, higher Doppler velocities, and warmer temperatures ([Garrett and Yuter 2014](#); [Kneifel and Moiseev 2020](#)), all properties of AR snowfall compared to NoAR snowfall found in this study (e.g., [Table 2](#); [Fig. S2](#)). [Moiseev et al. \(2017\)](#) further demonstrated that a higher fraction of riming corresponds with denser particles. The median AR effective density distributions are about $0.01\text{--}0.02 \text{ g cm}^{-3}$ denser than the median NoAR effective density distributions ([Figs. S3c,f](#)), which corresponds to about a 0.1 (i.e., 10%) increase in the fraction of riming ([Moiseev et al. 2017](#); their [Fig. 1](#)). However, matching the median fall speed distributions and effective density distributions to empirical relations does not provide definitive answers as to which ice habits and processes are present during AR and NoAR snowfall. While riming may contribute to the bulk microphysical differences observed during AR snowfall, we cannot provide a definitive answer and instead present it as a potential factor.

5. Conclusions

In this study, we applied a snowfall classification scheme to identify AR and NoAR snowfall events at instrument sites in Marquette, Michigan, and Hyttälä, Finland. We evaluated the large-scale, environmental, and the macro- and microphysical properties of these events to characterize AR snowfall and lend insight to potential processes. By examining the composite properties of AR and NoAR snowfall, we were able to determine the preferential synoptic setups, typical environmental conditions, and the characteristic snow particle properties of AR and NoAR snowfall.

We first evaluated the ERA5 large-scale properties of the events and found that AR snowfall has a preferential synoptic setup that transports enhanced moisture to both instrument sites. In MQT, the preferential synoptic setup was characterized by low pressure over the Upper Great Lakes that worked in conjunction with high pressure in the east to funnel enhanced moisture and warm temperatures from the Gulf of Mexico and

southern United States toward the site. In HYY, the preferential synoptic setup was characterized by a large region of low pressure over the northern Atlantic Ocean that combined with high pressure to the northwest of Africa to produce a tight pressure gradient and strong 10-m winds which funneled moisture and warm temperatures toward the site. Conversely, the NoAR snowfall events were characterized by preferential synoptic setups that instead advected drier air to each of the instrument sites. We also discussed the potential preference for high pressure blocking during AR snowfall in MQT and an apparent Atlantic Trough and NAJ extension weather pattern during AR snowfall in HYY.

After evaluating the large-scale properties of AR snowfall, we examined composite ERA5 temperature and specific humidity profiles near the sites and found that the AR snowfall events had significantly warmer and moister columns than the NoAR snowfall events. At both sites, temperatures during AR snowfall were about 2°–5°C warmer and about 0.4–0.7 g kg^{−1} moister throughout the column than NoAR snowfall.

To assess the impacts of these large-scale and environmental properties on the macro- and microphysical characteristics of AR snowfall, we explored the radar reflectivity characteristics, PSDs, fall speeds, and effective densities during the regimes. The 2D reflectivity histograms revealed that the AR snowfall events had higher reflectivity values throughout the column compared to the NoAR snowfall events. By inspecting the PIP PSDs, we concluded that AR snowfall events produce 38%–47% more particles than NoAR snowfall events in MQT and HYY, respectively. Similarly, by focusing on the PIP fall speed and effective density distributions, we found that AR snowfall events produce particles that fall faster and are denser than the NoAR snowfall events. These results were consistent across both sites and confirm that AR snowfall is more intense than NoAR snowfall.

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Data availability statement. ERA5 data are available for download via the Copernicus Climate Data Store (CDS) at the following link: <https://cds.climate.copernicus.eu/>. The data from the Hyytiälä, Finland, site used in this study are generated by the Aerosol, Clouds and Trace Gases Research

Infrastructure (ACTRIS) and are available from the ACTRIS Data Centre using the following link: <https://cloudnet.fmi.fi/site/hyytiäla>. The PIP and surface meteorological data used in this study are available on the University of Michigan Deep Blue Data repository (King et al. 2024a) and can be downloaded from <https://doi.org/10.7302/37yx-9q53>. The MRR data used in this study are available on the University of Michigan Deep Blue Data repository (Richter and Pettersen 2024) and can be downloaded from <https://doi.org/10.7302/4qddq-8184>.

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